# E-Waste (Management) Rules, 2022

Comparison of Five Electronics Manufacturers in India — With Implications

**Basis:** CPCB's FAQ on the E-Waste (Management) Rules, 2022 and amendments, together with the company disclosures reviewed for this assignment. The CPCB states that the 2022 Rules include EPR, registered-recycler requirements, EPR certificates, returns, audits and environmental-compensation provisions.

**Important:** “Rules not followed / not publicly evidenced” means the reviewed public material did not clearly demonstrate the requirement. It is not, by itself, proof of a legal violation.

## Key Rules Used for Comparison

**1. EPR registration:** Covered producers must register on the E-Waste EPR Portal.

**2. EPR recycling targets:** 60% for FY 2023-24 & 2024-25; 70% for FY 2025-26 & 2026-27; 80% from FY 2027-28 onwards.

**3. Registered recyclers:** EPR obligations must be fulfilled through registered e-waste recyclers.

**4. Environmentally sound management:** E-waste must be managed to protect health and the environment.

**5. RoHS:** New covered equipment/components must remain within specified hazardous-substance limits.

**6. Quarterly & annual returns:** Producers are required to submit periodic returns.

**7. EPR certificates:** Producers fulfil obligations by purchasing EPR certificates from eligible registered recyclers through the portal.

**8. Waste channelisation:** E-waste generated by bulk consumers is to be handed to registered producers/recyclers/refurbishers.

**9. Records & audit:** Stakeholders must maintain required records and make them available for verification/audit.

**10. Recycling residue:** Unrecycled fractions should go to appropriate registered recyclers; residues to authorised treatment/storage/disposal facilities.

**11. Registration documentation:** RoHS self-declaration and availability of technical documents are required for applicable producers.

**12. Hazardous substances:** Limits include lead, mercury, cadmium, hexavalent chromium, PBB and PBDE

# Dixon Technologies (India) Limited

| Rules followed / practices evidenced | Rules not followed / not publicly evidenced |
|---|---|
| • Uses government-authorised recyclers / agencies for e-waste, plastic and hazardous-waste handling.<br>• Reports a structured 3R approach: Reduce, Reuse and Recycle.<br>• Reports Zero Waste to Landfill (ZWL) across manufacturing facilities.<br>• States that waste disposal agencies approved by Central/State Pollution Control Boards are used.<br>• States that EPR is applicable to relevant activities and that waste collection is aligned with regulatory requirements.<br>• Reports restrictions on hazardous substances such as lead, mercury, cadmium and hexavalent chromium.<br>• Uses regulated channels for handling, storage, transportation and disposal of hazardous materials.<br>• As a B2B EMS manufacturer, explains that direct end-of-life product reclamation is generally not applicable to its manufacturing model. | • Current e-waste-specific EPR registration details under the 2022 Rules are not clearly disclosed in the material reviewed.<br>• Exact e-waste EPR recycling percentage achieved against the current 70% target is not publicly stated in the reviewed material.<br>• Quarterly and annual e-waste EPR return figures are not clearly presented in the reviewed public material.<br>• EPR certificate purchase/transaction details are not publicly stated in the reviewed material.<br>• A detailed public product take-back/reclamation programme is not reported; Dixon explains this through its B2B model.<br>• Current RoHS technical-document availability/self-declaration under the 2022 Rules is not clearly disclosed in the reviewed material. |

## Implications

| Area | Implication |
|---|---|
| Environmental | Authorised recycling and Zero Waste to Landfill practices can reduce uncontrolled disposal and improve material recovery. |
| Compliance | If applicable EPR targets or reporting obligations are not demonstrably met, regulatory scrutiny and environmental-compensation exposure may arise. |
| Operational | Maintaining authorised recycler relationships and traceable records can reduce waste-handling and audit risk. |
| Reputation | Publishing clearer EPR targets, certificates and return data would make its compliance position easier for customers and regulators to verify. |

# Kaynes Technology India Limited

| Rules followed / practices evidenced | Rules not followed / not publicly evidenced |
|---|---|
| • E-waste generated from operations is collected, stored and disposed of in accordance with the E-Waste Management Rules.<br>• Collaborates with CPCB-authorised recyclers and registered agencies for recycling, refurbishment and disposal.<br>• States that EPR is applicable and that waste collection/disposal practices are aligned with EPR plans submitted to Pollution Control Boards.<br>• Reports statutory environmental returns, including Form 3 (E-Waste Management) in its recent sustainability reporting.<br>• Uses source segregation and a Reduce-Reuse-Recycle approach for waste management.<br>• Hazardous waste is channelled through authorised facilities/vendors.<br>• FY 2024-25 reporting records 12.32 MT of e-waste generated.<br>• Reports use of RoHS-compliant SMT machinery and related environmental improvements. | • The exact quantity of EPR certificates purchased or generated is not disclosed in the reviewed material.<br>• The exact percentage of the statutory e-waste EPR recycling target achieved is not clearly disclosed.<br>• A detailed public consumer take-back network for finished products is not clearly described.<br>• Current RoHS self-declaration and technical-document availability under Rule 16 are not clearly disclosed in the reviewed material.<br>• Publicly available material does not clearly provide a product-wise EPR target/recycling reconciliation. |

## Implications

| Area | Implication |
|---|---|
| Environmental | Controlled collection and authorised recycling reduce the risk of e-waste entering unsafe disposal channels. |
| Compliance | Without exact EPR target and certificate data, external stakeholders cannot easily verify performance against the statutory obligation. |
| Operational | Continued use of authorised recyclers supports traceability and easier audit preparation. |
| Business | Clearer EPR metrics could strengthen ESG credibility and customer confidence. |

# Sahasra Electronics Private Limited

| Rules followed / practices evidenced | Rules not followed / not publicly evidenced |
|---|---|
| • Has publicly displayed an EPR Authorisation Certificate.<br>• Has an authorisation certificate for E-Waste Management.<br>• Partners with E-Waste Recyclers India, described as a recycler/dismantler registered with the Haryana Pollution Control Board.<br>• Provides a mechanism for collection of e-waste across India through collection points/reverse logistics.<br>• Provides customer instructions to hand over old electronics through the authorised collection/recycling channel.<br>• Uses an established recycler partner for dismantling/recycling at an authorised facility.<br>• Maintains ISO 14001 environmental-management certification information on its website.<br>• Publicly provides collection-centre information supporting organised channelisation. | • Its public e-waste page reviewed still prominently cites the older E-Waste (Management & Handling) Rules, 2016/2018 rather than clearly explaining the 2022 Rules.<br>• The current statutory 70% EPR recycling target achievement is not publicly quantified in the material reviewed.<br>• Quarterly and annual EPR return figures are not clearly disclosed on the public e-waste page reviewed.<br>• Current EPR certificate transactions/purchases are not publicly disclosed.<br>• Current Rule 16 RoHS declarations and technical-document availability are not clearly disclosed.<br>• A current product-wise reconciliation against the 2022 EPR targets is not publicly shown. |

## Implications

| Area | Implication |
|---|---|
| Environmental | Authorised collection and recycling provide a formal route for end-of-life electronics and reduce the risk of unsafe informal processing. |
| Compliance | Use of older-rule terminology on a public page can create ambiguity about tracking and reporting under the 2022 Rules. |
| Operational | A nationwide collection/recycler network can improve reverse logistics and traceability. |
| Reputation | Updating public disclosures to the current Rules and publishing measurable EPR results would improve transparency. |

# Syrma SGS Technology Limited

| Rules followed / practices evidenced | Rules not followed / not publicly evidenced |
|---|---|
| • Has an authorised recycler/dismantler partner responsible for collection, transportation, storage, segregation, dismantling and recycling of e-waste.<br>• Provides authorised e-waste collection points and a toll-free disposal channel.<br>• Explicitly advises customers not to give e-waste to informal/unorganised scrap channels.<br>• Directs end-of-life electronics to authorised collection/recycling points.<br>• FY 2024-25 reporting states use of lead-free, RoHS- and REACH-compliant solder paste and bars.<br>• Waste management is stated to be implemented in line with ISO 14001 principles.<br>• Reports categorisation of waste and monitoring of waste-management performance.<br>• Its annual reporting provides detailed waste-reclamation disclosures and explains its B2B EMS model. | • Exact e-waste EPR recycling-target achievement is not clearly stated in the reviewed public material.<br>• EPR certificate purchase/transaction details are not publicly disclosed.<br>• Quarterly and annual EPR portal return figures are not clearly presented.<br>• The company states it is a B2B contract manufacturer and does not control end-of-life handling of finished goods, so a consumer take-back programme for finished products is not reported.<br>• Current Rule 16 producer-registration/RoHS documentation details are not clearly presented on the public EPR page. |

## Implications

| Area | Implication |
|---|---|
| Environmental | Authorised collection and recycling reduce the probability of unsafe dismantling and improve material recovery. |
| Compliance | RoHS/REACH-oriented material controls support hazardous-substance management, but measurable EPR performance still needs public evidence. |
| Operational | A dedicated recycler/dismantler simplifies collection, transportation and recycling traceability. |
| Business | As a B2B contract manufacturer, clearly separating manufacturing waste from end-of-life customer products is important for stakeholder understanding. |

# Centum Electronics Limited

| Rules followed / practices evidenced | Rules not followed / not publicly evidenced |
|---|---|
| • States that e-waste is disposed of through an approved e-waste recycler.<br>• Uses a waste-management system involving collection, segregation, storage and disposal.<br>• Uses authorised vendors/recyclers for waste streams.<br>• Hazardous waste is disposed through approved hazardous-waste incinerators.<br>• Plastic waste is routed through an approved plastics recycler.<br>• Reports a Life Cycle Assessment (LCA) process for products.<br>• FY 2024-25 report records 6.30 MT of e-waste generated.<br>• Reports recovery/recycling and reuse data in its waste-management disclosures.<br>• Maintains annual reports and annual-return disclosures on its investor-relations site. | • The FY 2024-25 report states that EPR is 'Not Applicable' to the entity's activities; therefore, an e-waste EPR target/registration is not demonstrated in that report.<br>• No current e-waste EPR recycling percentage is reported against the statutory producer targets in the reviewed material.<br>• EPR certificate purchase/transaction details are not disclosed.<br>• Quarterly/annual e-waste EPR portal return details are not clearly disclosed.<br>• Current Rule 16 RoHS self-declaration and technical-document availability are not clearly disclosed.<br>• A consumer-facing e-waste take-back programme for finished products is not clearly described in the reviewed report. |

## Implications

| Area | Implication |
|---|---|
| Environmental | Approved recycling and controlled waste handling reduce risks from improper disposal of e-waste and hazardous waste. |
| Compliance | Since the FY 2024-25 report states EPR is not applicable, that applicability assessment should remain clearly supported by the company's business/activity scope. |
| Operational | Documented waste quantities and approved vendors improve traceability and audit readiness. |
| Reputation | A clear explanation of EPR non-applicability and the company's e-waste pathway would reduce ambiguity for stakeholders. |

## Overall Implications

| Area | Implication |
|---|---|
| Environmental | Effective EPR and authorised recycling reduce the risk of uncontrolled e-waste disposal and support recovery of useful materials. |
| Compliance | Failure to meet applicable obligations can expose a producer to regulatory scrutiny and environmental compensation; the CPCB FAQ explicitly provides for environmental compensation in case of violations. |
| Financial | Producers may need EPR certificates to fulfil obligations, so insufficient recycling performance can create additional compliance costs. |
| Operational | Authorised recycler contracts, collection systems, records and audit-ready documentation improve traceability and reduce waste-handling risk. |
| Reputation | Transparent reporting of EPR registration, recycling targets, certificates and returns can strengthen ESG credibility and stakeholder confidence. |

## Conclusion

Thefive companies show different levels of publicly evidenced compliance. A common strength is controlled waste handling through authorised recyclers, while a common disclosure gap is detailed evidence of current EPR performance under the 2022 Rules. The implications therefore cover environmental protection, compliance, operational traceability, potential cost and stakeholder confidence.